

Research Directions for Modal Boundary-Integral Inverse Scattering

Executive verdict

The research question is now sufficiently specified by the supplied project brief: **which numerical-analysis direction should be pursued next for a two-dimensional frequency-domain TMz dielectric inverse-scattering code based on a Müller boundary-integral formulation, Fourier/Laurent geometry, modal traces, and repeated multi-frequency geometry updates?** The target audience appears to be a researcher in numerical analysis/inverse scattering; the target journal, project deadline, available compute budget, and current completeness of the modal prototype are unspecified. I therefore treat every proposed non-Nyström capability as unproven until validated against the existing Kress–Nyström reference. [filecite turn0file0](#)

My recommendation is to make the main research direction a derivative-aware hybrid Fourier–Galerkin Müller method: analytically isolate the singular Fourier structure, numerically represent only the smooth modal remainder, and compress the forward operator and its shape derivatives together. The key novelty should *not* be “Fourier–Galerkin,” “logarithmic splitting,” “Maue’s identity,” or “Laurent geometry.” All have substantial antecedents. The publishable question is instead whether, for smooth closed dielectric interfaces, the **desingularized Müller operator and its geometry derivatives have exploitable modal coupling structure that can be predicted or adaptively detected, retained without degrading inverse gradients, and converted into an accuracy-matched end-to-end saving.** Classical trigonometric BIE discretization and logarithmic product quadratures are well established; Fourier–Galerkin transmission methods and Fourier-matrix compression also already exist. ¹ What appears substantially less covered is their combination into a derivative-aware modal compression mechanism for a closed-interface Müller transmission system used in geometry inversion. This is an inference from the literature search, not a claim of priority.

The **strongest alternative** is a **local joint geometry–frequency operator surrogate**. Around the current inverse iterate c_0 , approximate the Müller matrix and its geometry derivatives simultaneously in shape and frequency, while preserving a distinct linear solve at every physical frequency. This has potentially greater practical payoff than modal compression because assembly and derivative work could be amortized over frequencies and line-search/trust-region trials. But its novelty is more threatened: frequency-decoupled BEM/MOR, electromagnetic multi-frequency shape optimization, broadband BEM sensitivity, high-order transmission shape Taylor expansions, and two-dimensional electromagnetic shape sensitivities have all advanced materially through 2024–2026. ² A paper would therefore need a genuinely new **joint, gradient-controlled reuse mechanism**, not merely Taylor expansion of Hankel functions plus shape differentiation.

A **lower-risk alternative** is inverse-aware reduced-basis/Krylov recycling across frequency and nearby geometry. It is an excellent engineering diagnostic and could become part of either leading method, but I would not currently make generic reduced-basis acceleration the main publication: electromagnetic RB-BEM, dielectric RB-BEM, greedy multi-frequency acoustic bases, and multiparameter acoustic Krylov recycling already cover much of that conceptual territory. ³

The **most tempting direction to defer is a completely “node-free” direct Laurent-coefficient implementation of every Müller block**. It is arguably the most mathematically elegant candidate, and potentially the most theoretically interesting, but it combines the highest implementation risk with difficult validation and unclear end-to-end payoff. Related coefficient-domain layer-potential representations already exist when genuine conformal mappings are available, so “operator algebra from boundary coefficients” is not itself unprecedented. ⁴ More importantly, a general Laurent boundary parametrization $Z(e^{it}) = \sum c_p e^{ipt}$ is not automatically the boundary trace of the particular univalent conformal map required by such theories. The hybrid experiment below can determine whether the useful modal structure exists **before** committing to exact coefficient algebra.

The smallest experiment I would run next requires no new singular quadrature at all: use the existing Nyström code to generate accurately converged matrices on a few noncircular shapes, Fourier-project them, finite-difference the matrices in a few geometry directions, and ask whether both the modal matrices **and their directional shape derivatives** can be aggressively truncated while preserving receiver fields and directional objective gradients. If that test fails, a large fraction of the motivation for both hybrid modal compression and fully direct Laurent assembly disappears.

Candidate comparison and Pareto view

Scores use a five-point scale. For **novelty, theory, and payoff**, larger is better. For **implementation difficulty and validation difficulty**, larger means harder. The scores are deliberately not combined into a weighted total.

Candidate direction	Separate scores: novelty / implementation / validation / theory / payoff	Closest established precedent	Plausible paper- level contribution	Principal failure risk
Derivative-aware hybrid modal Müller compression — recommended	4.2 / 3.2 / 3.2 / 4.7 / 4.2	Trigonometric/ log-singular BIEs; Fourier- Galerkin transmission; $O(n \log n)$ Fourier matrix compression. ¹	A compression/ truncation law for all Müller blocks and their geometry derivatives , with forward/ gradient error control and inverse- scattering validation	Helmholtz mode coupling or derivative matrices become essentially dense as ka , shape complexity, or geometry bandwidth grows

Candidate direction	Separate scores: novelty / implementation / validation / theory / payoff	Closest established precedent	Plausible paper- level contribution	Principal failure risk
Joint geometry- frequency operator/ derivative surrogate — strongest alternative	3.8 / 3.6 / 4.1 / 4.4 / 5.0	Frequency extraction, EM frequency- decoupled MOR/ shape optimization, broadband sensitivity, high- order shape Taylor. ⁵	Adaptive local separated representation of $A(c, \omega)$ and $\partial_c A$ whose acceptance/ rebuild rule is based on inverse-gradient error	Required shape radius is smaller than one useful optimization step, or frequency rank/ subdivision destroys amortization
Inverse-aware reduced basis / Krylov recycling	2.4 / 2.4 / 3.0 / 2.8 / 4.0	EM EFIE RBM, dielectric RB- BEM, acoustic frequency RB and multiparameter Krylov. ⁶	A geometry- changing basis/ recycling policy whose error indicator guarantees useful receiver and adjoint- gradient accuracy	Assembly remains dominant, or the solution manifold changes enough that U must be rebuilt almost every geometry step
Nearby-frequency/ geometry preconditioner or approximate- inverse reuse	2.5 / 2.7 / 2.7 / 3.0 / 4.0	BIE solvers for locally perturbed geometries reuse an existing direct solver via low-rank updates. ⁷	Show that global low- bandwidth Fourier shape updates induce a practically reusable preconditioned operator even though they are not geometrically local	Global Fourier perturbations are not low- rank matrix perturbations; update quality collapses rapidly

Candidate direction	Separate scores: novelty / implementation / validation / theory / payoff	Closest established precedent	Plausible paper- level contribution	Principal failure risk
Fully direct Laurent coefficient assembly — defer	3.8 / 5.0 / 4.7 / 5.0 / 3.5	Conformal/ coefficient- domain layer- potential matrices exist in related planar Helmholtz/ Laplacian problems. ⁴	Stable coefficient-only Müller assembly with rigorous truncation/error accounting and competitive end-to-end complexity	Intermediate bandwidth explosion, aliasing, cancellation, hypersingular complexity, and no speed advantage over an FFT hybrid
Geometry-adapted Fourier basis / pseudodifferential modal preconditioner	3.0 / 3.2 / 3.1 / 4.2 / 3.5	On smooth domains, acoustic boundary operators admit approximate descriptions in transplanted circle Fourier modes and a normal-plus- smooth- perturbation structure. ⁸	Connect geometry- dependent mode coupling to an adaptive basis/ preconditioner and observable iteration reductions	Interesting spectral structure does not translate into enough total-cost improvement for modest M
Multi-object local scattering matrices with translation operators	2.0 / 3.5 / 3.1 / 3.1 / 4.0 once multi-object physics is needed	Arbitrary smooth particles can be represented by reusable local scattering matrices coupled through multipole translations. ⁹	Inverse updates that reuse local object response while positions/ shapes change	It expands the physical scope before the single-object modal method is settled; much of the core forward mechanism is classical

Candidate direction	Separate scores: novelty / implementation / validation / theory / payoff	Closest established precedent	Plausible paper- level contribution	Principal failure risk
Time-domain / convolution- quadrature backend	1.8 / 5.0 / 4.2 / 3.5 / 2.0 for present scope	High-order Nyström/CQ methods already solve time- domain scattering via families of complex- frequency Helmholtz BIEs. 10	Would require a distinct broadband inverse advantage, not merely another route to many frequencies	Major backend change ultimately recreates a large set of frequency- domain BIE solves and distracts from the current inverse variables

Pareto interpretation. If the priority is **credible publication potential at tolerable risk**, derivative-aware modal compression is the best balance. If the priority is **maximum eventual end-to-end inverse speed**, the joint geometry-frequency surrogate has the highest upside, but also the most serious novelty and validation burden. If the priority is **implementation safety**, reduced-basis/Krylov or preconditioner recycling wins, but the likely methodological novelty is much lower. If the priority is **pure mathematical elegance**, fully direct Laurent assembly is the most attractive; that is exactly why it is dangerous as the first investment.

A particularly important negative conclusion is that **smaller algebraic systems are not automatically a lower-cost algorithm**. A reduced solution space may leave matrix assembly untouched; a compressed matrix may reduce memory but not dense factorization cost; a direct coefficient representation may replace kernel evaluation by expensive coefficient convolutions; and an approximation that reproduces fields can still corrupt geometry gradients. Existing RB work demonstrates the first distinction particularly clearly: RB methods accelerate many-query solves by projecting onto a low-dimensional solution space, whereas frequency-extraction methods attack the frequency dependence of the BEM matrix itself. 11

Main direction: derivative-aware hybrid Fourier–Galerkin Müller compression

For a trace bandwidth M , let $A_M(c, \omega) \in \mathbb{C}^{2M \times 2M}$ denote the **true Fourier–Galerkin** discretization of the pulled-back Müller system, not merely $P^* A_N P$. The boundary Fourier coefficients are $c = (c_p)$, and ω is a physical angular frequency. For each integral block, first make the standard two-dimensional split

$$\kappa(t, \tau; c, \omega) = a(t, \tau; c, \omega) \ell(t - \tau) + b(t, \tau; c, \omega),$$

with

$$\ell(\theta) = \log \left(4 \sin^2 \frac{\theta}{2} \right), \quad \hat{\ell}_j = -\frac{1}{|j|} (j \neq 0), \quad \hat{\ell}_0 = 0.$$

The singular Fourier coefficients are therefore known analytically; the geometry- and frequency-dependent amplitudes a and the desingularized remainder b are smooth under the appropriate regularity assumptions. Trigonometric discretization, singularity extraction, and spectrally accurate treatment of smooth two-dimensional Helmholtz boundaries have a long history, so this split cannot be presented as new. ¹²

The proposed new object is instead a **selected modal representation**

$$\tilde{A}_M(c, \omega) = A_M^{\text{sing}}(c, \omega) + \Pi_{\mathcal{S}(c, \omega)} R_M(c, \omega),$$

where R_M is the modal matrix of the desingularized remainder and $\Pi_{\mathcal{S}}$ retains an adaptively chosen set $\mathcal{S}$ of input/output mode couplings. The essential extension for inverse scattering is

$$\partial_{c_j} \tilde{A}_M = \partial_{c_j} A_M^{\text{sing}} + \Pi_{\mathcal{S}_j} \partial_{c_j} R_M,$$

with truncation chosen to control **derivative error**, not merely the forward operator. Here c_j is one real geometry parameter—for complex Cartesian Fourier coefficients, it is cleaner computationally to differentiate with respect to their real independent components.

If the desingularized kernel extends analytically to a complex strip in both periodic variables, its two-dimensional Fourier coefficients have exponential decay; with only finite smoothness the decay becomes algebraic. This general Fourier principle is the mathematical basis of existing Fourier–Galerkin compression, where Cai and Xu obtained $O(n \log n)$ retained coefficients for a class of smooth compact integral kernels. ¹³ The nontrivial work here would be to establish what survives for **all four coupled transmission blocks, the hypersingular block after regularization, and shape derivatives**. Betcke, Phillips and Spence provide complementary motivation: on general smooth boundaries, acoustic layer operators can be understood as a simpler approximately Fourier-diagonal/normal component plus a smooth perturbation, although that result does **not** imply that an arbitrary Müller matrix will be strongly banded in the chosen parameter t . ⁸

The intuition is simple: **the diagonal singularity has a universal, highly structured Fourier representation; all geometrically complicated mode mixing is moved into smoother functions. The research question is how much of that remaining mixing really needs to be retained, and whether the same economy survives differentiation with respect to shape.**

This distinction matters because Fourier spectral BIEs are already substantial prior art. Pinto, Aylwin and Jerez-Hanckes developed a rigorous Fourier spectral Galerkin method for two-dimensional Helmholtz transmission in periodic multilayer gratings, with super-algebraic convergence and performance competitive with Nyström discretization. ¹⁴ Cai and Xu already compressed Fourier matrices, and trigonometric Galerkin treatments of hypersingular equations with matrix truncation also predate the proposed project. ¹⁵ Thus the proposed paper cannot credibly be “a fast Fourier–Galerkin method for transmission.”

A defensible paper claim would instead look like:

For smooth closed dielectric interfaces, derive and validate a derivative-aware compressed Fourier–Galerkin Müller discretization in which the singular modal interactions are treated analytically and the desingularized forward and shape-derivative matrices are adaptively compressed, with explicit forward/gradient error criteria and demonstrable accuracy-matched savings in geometry inversion.

The strongest theoretical version would relate the retained coupling set to geometry regularity, wavenumber, and trace bandwidth. A weaker but still potentially publishable algorithmic version would supply an adaptive coefficient-selection strategy with robust a posteriori error indicators and show that the *same compressed representation* supports reliable adjoint gradients.

The pipeline impact is unusually contained. **Geometry processing** still begins from $Z(t) = \sum c_p e^{ipt}$. **Assembly** changes substantially. **Factorization and field solve** can initially remain conventional dense operations so that assembly is studied independently; only later should sparsity-aware iterative/direct solvers be introduced. **Receiver evaluation** need not change. **Derivatives** change substantially because compressed $\partial A / \partial c_j$ becomes a first-class numerical object. **Optimization** should initially remain identical to the reference inverse method so that any observed reconstruction difference is attributable to the forward/gradient approximation.

There is also a useful intermediate interpretation of “node-free.” This recommended method is **not node-free**. It may evaluate the smooth remainder on an oversampled (t, τ) grid and transform it with FFTs, or sample selected coefficient regions adaptively. What it can avoid is **singular boundary-pair quadrature for the analytically extracted part**. That is a much more precise and defensible claim than “no nodes.” A later fully coefficient-domain implementation would need to distinguish three progressively stronger statements: no singular quadrature, no boundary-point-pair kernel evaluation, and no physical-space sampling anywhere. Those are not equivalent.

Why not go immediately to direct Laurent algebra? Products, quotients, norms, normal vectors, Bessel/Hankel series and differentiated kernels all enlarge intermediate coefficient bandwidth. Multiplication gives convolutional bandwidth growth even before nonlinear analytic functions are approximated. Truncation introduced between those operations creates aliasing and cancellation questions, while high k generally increases the number of oscillatory modes that must be represented. In addition, related coefficient-matrix constructions based on conformal maps assume mathematical structure stronger than “the boundary happens to have a Laurent parametrization.”⁴ The hybrid method is therefore a strategically useful testbed: it can reveal the actual modal coupling pattern without first solving all coefficient-algebra problems.

Novelty confidence: medium-high, conditional on derivative-aware compression and a real mathematical or adaptive truncation result. I did not find a paper that already supplies this complete mechanism for a closed two-medium Müller system and then verifies inverse geometry gradients. That unsuccessful search is **not** proof of priority. A broader MathSciNet/Scopus citation-chain search should be done before manuscript positioning.

What would make the work unpublishably incremental is clear: implementing the known logarithmic Fourier weights, doing a two-dimensional FFT of a smooth remainder, presenting only circular-cylinder tests, or reporting raw assembly time without showing a new compression mechanism and gradient preservation would be too close to established spectral BIE technology.¹⁶

Strong alternative: a joint geometry–frequency operator surrogate

The second direction asks a different question: can work performed at one geometry iterate and a few frequencies be reused **across both frequency and nearby shape changes**?

A useful local representation is

$$A(c, \omega) \approx \sum_{r=1}^R f_r(\omega) \left[A_r^{(0)} + \sum_j \delta c_j A_{r,j}^{(1)} + \frac{1}{2} \sum_{j,k} \delta c_j \delta c_k A_{r,jk}^{(2)} + \dots \right],$$

where

$$\delta c = c - c_0,$$

c_0 is the current geometry anchor, $f_r(\omega)$ are chosen frequency basis functions, and $A_{r,j}^{(1)}$, $A_{r,jk}^{(2)}$ are local geometry derivative coefficient matrices. The frequency representation could be a Taylor basis only on sufficiently small intervals, but Chebyshev, rational, adaptive cross, or other separated approximations are preferable candidates for wider bands. The representation is a **truncated approximation**, not an identity.

Crucially, it does **not** turn all frequencies into one physical system. For every illumination frequency,

$$\tilde{A}(c, \omega_\ell) x_\ell = b(c, \omega_\ell)$$

is still solved separately. What is shared is expensive matrix/derivative preparation. This respects the project brief's distinction between legitimate cross-frequency reuse and invalidly adding frequency matrices. [filecite turn0file0](#)

For a real inverse objective

$$L(c) = \sum_{\ell} w_{\ell} L_{\ell}(c),$$

an adjoint calculation contains terms of the schematic form

$$\frac{\partial L}{\partial c_j} = \sum_{\ell} \text{Re} \left[\lambda_{\ell}^* \left(\frac{\partial b_{\ell}}{\partial c_j} - \frac{\partial A_{\ell}}{\partial c_j} x_{\ell} \right) \right] + \text{explicit receiver/regularization terms.}$$

Therefore a surrogate suitable for inversion needs accurate $\partial A / \partial c_j$, not merely small field error. Shape differentiability of electromagnetic boundary-integral operators is well established: Costabel and Le Louër proved infinite differentiability for broad classes of BIE operators and treated singular/hypersingular operators via weakly singular operators and surface differential operators. ¹⁷ More recently, Bao and collaborators developed high-order shape Taylor recurrence formulas covering electromagnetic and transmission scattering, and their 2026 two-dimensional numerical paper explicitly treats transmission conditions. ¹⁸ Thus “we derive a shape derivative” is no longer a sufficiently strong novelty claim.

Frequency approximation is equally crowded. Dirckx, Huybrechs and Meerbergen developed frequency extraction for three-dimensional scalar Helmholtz BEM matrices, separating oscillatory frequency dependence and approximating frequency families in data-sparse form. ¹⁹ Chen et al. combined series-based frequency decoupling, SOAR model reduction and isogeometric electromagnetic multi-frequency shape optimization in 2024. ²⁰ In acoustics, 2025 work combines series-expanded frequency dependence, SOAR and direct shape sensitivity equations. ²¹

The electromagnetic overlap has become even closer. A 2025 paper treats two-dimensional dielectric TE scattering with IGABEM and analytically derived shape sensitivities validated against finite differences. ²² A September 2026 paper treats two-dimensional TM-polarized electromagnetic BEM sensitivity with analytical scattered-field and RCS derivatives with respect to geometry and incidence direction. ²³

Separately, two-dimensional dielectric multi-frequency shape optimization with NURBS-based BEM had already been demonstrated in 2024. ²⁴

That recent literature changes the novelty target considerably. **The contribution cannot merely be “multi-frequency EM + shape sensitivity.”**

A defensible contribution would instead be:

An adaptive, local, joint geometry-frequency surrogate for a dielectric Müller operator and its adjoint-relevant shape derivatives, equipped with a trust-region/rebuild criterion based on receiver and gradient error, and shown to reduce the total cost of actual multi-frequency geometry reconstruction.

That is an **algorithmic mechanism/integration contribution**, unless accompanied by new approximation-rank or truncation theorems. The central conceptual freshness is that the surrogate is built around *what an inverse optimizer actually reuses*: one current geometry, many frequencies, multiple illuminations, and several nearby trial steps. It is neither generic fixed-geometry frequency sweeping nor a global solution ROM.

The reusable part is the local set of geometry/frequency coefficient matrices and perhaps their factorizations/preconditioners. Frequency-specific work still includes scalar frequency evaluation, right-hand sides, one system solution for each ω_ℓ , and receiver fields. Geometry-specific rebuilding remains necessary when the step leaves the local validity region.

This should help only if the inverse trajectory is locally smooth enough that several evaluations occur within one surrogate region, and if the frequency rank R remains modest. The matrix entries may vary smoothly with frequency even when the **solution** varies sharply near resonant or poorly conditioned regimes; smooth parameter dependence therefore does not prove low solution rank. That distinction is one reason to compress the operator first and treat solution reduction as optional.

The most concrete failure mode is fatal and easy to test: the geometry validity radius may be smaller than a useful optimizer step. If a new surrogate has to be constructed at essentially every candidate geometry, “reusable preparation” becomes merely a more expensive form of ordinary assembly.

Novelty confidence: medium. The 2024–2026 literature is close enough that the exact mechanism and error-control story must be sharp. Without a gradient-based acceptance/rebuild rule or a genuinely new separated derivative representation, I would judge this direction too incremental for a strong numerical-analysis paper.

The following comparison makes the overlap explicit.

Closest precedent	Physical/numerical setting	Frequency and geometry treatment	Demonstrated benefit	What would still have to be new here
Dirckx–Huybrechs–Meerbergen, 2022	3D scalar Helmholtz BEM; fixed geometry	Frequency extraction/data-sparse approximation; no moving geometry	Reduces cost of matrix families across frequency. ¹⁹	Geometry-dependent derivative reuse and inverse-update acceptance
Chen et al., 2024	Electromagnetic IGABEM, PEC formulation in the paper's description; NURBS	Taylor-type frequency decoupling + SOAR; multi-frequency shape optimization	Efficient broadband BEM/MOR integrated into shape optimization. ²⁰	Dielectric Müller transmission, inverse reconstruction, gradient-consistent joint shape/frequency reuse
Li et al., 2025	Acoustic BEM with ground reflection	Frequency series + SOAR + directly differentiated shape-sensitivity BIE	Broadband sensitivity acceleration. ²⁵	A new joint surrogate mechanism rather than transferring the same recipe to TMz
Bao–Lai–Ma, 2026	2D wave scattering incl. transmission	High-order shape Taylor; application demonstrated for UQ	High-order approximation under geometry perturbations. ²⁶	Coupling to frequency separation and inverse-gradient/error-controlled reuse
Shen et al., 2025; Guo et al., 2026	2D electromagnetic BIE sensitivity; dielectric TE in Shen et al., TM sensitivity in Guo et al.	Analytic geometry/current/RCS sensitivities	Validated geometry derivatives. ²⁷	Reusing derivatives across a frequency band and geometry trust region to accelerate an actual reconstruction

Lower-risk alternative: inverse-aware reduced solution spaces and recycling

A reduced solution representation takes a different form:

$$x(c, \omega) \approx Uz(c, \omega),$$

where $U \in \mathbb{C}^{2M \times r}$ contains $r \ll 2M$ solution-space basis vectors. A Galerkin reduced model would solve

$$U^* A(c, \omega) U z = U^* b(c, \omega).$$

This is **solution compression**. It should not be confused with the first direction, which compresses the modal operator, or the second direction, which approximates parameter dependence of the operator and its derivatives.

The appeal is implementation safety. Existing full matrices can be used unchanged, and snapshots can be generated with the current Kress–Nyström solver. Electromagnetic BEM reduced bases are already mature enough that this route has little conceptual uncertainty: Fares et al. introduced RBM for the EFIE in 2011; Shi et al. demonstrated a reduced-basis BEM for parameterized dielectric electromagnetic scatterers with reported online solution speedups of several to dozens of times at prescribed accuracy.

²⁸ Greedy multi-frequency reduced bases have likewise been demonstrated for acoustic BEM, including a reported example in which ten full solves covered 171 frequencies and reduced total solve-side wall time by more than 60% relative to frequencywise GMRES. ²⁹ Panagiotopoulos, Desmet and Deckers extended Krylov recycling to multiparameter acoustic BEM systems including non-affine parameter dependence. ³⁰

This density of prior art is exactly why I would use it first as a **diagnostic or component**, not as the headline research contribution.

A potentially stronger inverse-oriented variant would maintain U across nearby geometry iterates and enrich it only when a receiver or adjoint residual estimator fails. Ideally U would include information from both forward and adjoint solutions:

$$U = \text{span}\{x_\ell^{(k)}, \lambda_\ell^{(k)}, \dots\},$$

where k indexes selected geometry iterates and ℓ frequencies. That might allow one reduced space to preserve the quantities that actually enter the geometry gradient rather than simply minimizing a state-vector norm.

The idea is intuitive: nearby frequencies and nearby geometries often generate correlated boundary traces, so previously solved fields may span much of the new solution. The concrete failure mode is equally intuitive: as geometry changes, resonances move and new modal content appears. A basis that is tiny at fixed geometry can expand rapidly across a reconstruction trajectory.

A second limitation is more important for this project: **solution reduction does not remove full-order matrix assembly unless it is coupled to an affine/separated operator representation or hyper-reduction**. The classic RB literature obtains very cheap online stages when an effective offline/online decomposition is available. ³¹ If current timing shows that Müller assembly and shape-derivative assembly dominate factorization/solves, a reduced state dimension alone attacks the wrong bottleneck.

Nearby-frequency preconditioner recycling has similar status. Reusing a factorization or approximate inverse is attractive, and integral-equation solvers for *locally* perturbed geometries can exploit low-rank updates: Zhang and Gillman's method uses the fact that a local geometric change yields a low-rank update and reports roughly two- to three-fold improvements over rebuilding in their examples; their later extended-system formulation reports still larger precomputation reductions for its tested local perturbations. ⁷ But a global Fourier geometry update changes every quadrature interaction. There is no reason to assume that it has the same low-rank matrix-update structure. Testing singular values of

$$A(c_0 + \delta c, \omega) - A(c_0, \omega)$$

is cheap; assuming low rank without that test would not be justified.

The paper claim for reduced-basis work would need to be more specific than “RB accelerates Müller BIEs”:

An adaptive forward–adjoint reduced space shared across frequency and nearby shape iterates, with error indicators tied to receiver residuals and geometry gradients, reduces end-to-end inverse cost without changing the recovered geometry.

Even that claim faces substantial overlap with multiparameter MOR and electromagnetic shape optimization. ³² I therefore rate its standalone novelty as low-to-moderate but its value as an implementation control as high.

Novelty gaps and decisive experiments

What is already established versus what remains worth contributing

Ingredient	Already established	What this project would still need to contribute
Periodic Fourier/trigonometric treatment of smooth 2D BIEs and logarithmic singularities	Classical Nyström/Galerkin/trigonometric methods already handle these with rapid or exponential convergence for smooth/analytic data. ¹²	Nothing publishable merely from using Fourier modes or exact log coefficients
Fourier–Galerkin Helmholtz transmission	Rigorous fast spectral Galerkin transmission methods exist, including super-algebraic convergence in smooth periodic geometries. ¹⁴	Closed-interface Müller-specific compression and derivative/inverse consequences
Fourier coefficient compression	$O(n \log n)$ -type compressed Fourier matrices for suitable singular integral equations are established. ¹³	A structure/result covering the coupled Müller blocks, frequency dependence and shape derivatives
Shape differentiability of scattering BIEs	Infinite shape differentiability of broad electromagnetic BIE operators and high-order transmission shape-Taylor recurrences are established. ³³	Efficient derivative <i>representation and reuse</i> , plus gradient-error control
Multi-frequency BEM MOR/frequency extraction	Matrix-level frequency approximation, Krylov/RB sweeps and EM frequency-decoupled MOR exist. ³⁴	A mechanism genuinely adapted to changing geometry and inverse gradients

Ingredient	Already established	What this project would still need to contribute
2D electromagnetic geometry sensitivity	Dielectric TE sensitivities and 2D TM sensitivities have been published through 2025–September 2026. ²⁷	Broadband geometry-derivative reuse in a Müller inverse reconstruction, not another sensitivity derivation
Coefficient-domain layer-potential algebra	Infinite matrix formulations from conformal-map coefficients exist for related planar layer-potential/eigenvalue problems. ⁴	Stable arbitrary Fourier/Laurent boundary assembly for full Helmholtz transmission, with explicit truncation and complexity evidence
Multi-object scattering matrices	Reusable single-particle scattering responses coupled through local/multipole expansions are established. ³⁵	Only worth revisiting once multiple inclusions become part of the actual inverse problem

The decisive test for the recommended direction

Hypothesis. On representative smooth noncircular boundaries, a Fourier representation of the Müller operator contains a compressible smooth-coupling component, and—more importantly—the same is true of directional geometry derivatives at useful ka . Keeping only a modest fraction of modal couplings should preserve fields and geometry gradients.

Minimal code change. Do **not** implement direct coefficient assembly first. Use the converged Kress matrix already available. For each selected geometry and frequency, form a carefully weighted Fourier projection

$$A_M^{\text{proj}} = P^* A_N P$$

as a *diagnostic*. This is explicitly not being claimed to equal the eventual continuous Fourier–Galerkin matrix. Then compute directional matrix derivatives from two independently assembled geometries,

$$D_d A_M \approx \frac{A_M(c + hd) - A_M(c - hd)}{2h},$$

where d is a normalized Fourier-geometry perturbation. Inspect two-dimensional coefficient magnitude, numerical band profiles, singular values of off-band pieces, and truncation error. This experiment uses projected Nyström only to decide whether a more ambitious Galerkin assembler is justified.

Use at least a circle for analytic sanity, an ellipse, and one smooth noncircular geometry containing several positive and negative Laurent modes. Test low, intermediate, and moderately high electrical size; the exact ka range should be matched to the intended GPR problem, but a provisional spread such as $ka \approx 2, 5, 10$ is much more informative than a single low-frequency test. Geometry bandwidth, trace M , Nyström N , and any FFT oversampling must be recorded separately.

Controls. The circle should agree with cylindrical-wave analytics. Noncircular projected operators and fields should be checked against independently refined Kress runs. Matrix derivatives should be stable under halving h until discretization/roundoff error takes over. After truncating a matrix, solve the *actual*

transmission problem and evaluate receiver fields; do not judge compression from matrix-entry pictures alone.

Measurements. Record retained-entry fraction, operator-action error on random and physical vectors, receiver-field error, condition number or solve sensitivity, directional objective-gradient error, memory, and actual matrix-vector/assembly work. Repeat after doubling M .

Provisional go criterion. Continue if, on the noncircular cases rather than just the circle, a structured/adaptive retention set of roughly **30% or less of dense couplings** preserves receiver fields to about 10^{-6} relative error and directional inverse gradients to about 10^{-4} – 10^{-3} , and if the retained fraction does not trend rapidly toward one as M is doubled. Those numerical thresholds are research gates, not universal accuracy requirements; the relevant standard is that surrogate error should sit safely below the independently measured discretization and inversion-error budget.

Stop condition. Stop making modal compression the main research direction if the derivative matrices need more than roughly half of all modal interactions already at the intended moderate ka , or if compression deteriorates rapidly with trace refinement. In that case, a “fast modal” paper would be relying on a low-frequency/special-geometry effect.

If the diagnostic passes, only then build the true Fourier–Galerkin blocks: first V , then K/K' , and finally T through a carefully verified regularized/Maue route. The full Müller solve is the gate after the scalar-block mechanism has survived.

The decisive test for the joint geometry–frequency surrogate

Hypothesis. Over a frequency band and the size of shape perturbations actually proposed by an optimizer, the family

$$\{A(c, \omega), D_d A(c, \omega)\}$$

has sufficiently low joint numerical complexity that a few frequency anchors and low shape order predict held-out matrices and gradients.

Again, the decisive experiment can be done **without implementing the final surrogate**. Generate full Nyström matrices at a small tensor of shape perturbations and frequencies. Use one-dimensional shape directions first; this avoids the combinatorial cost of a multivariate Taylor tensor. Fit Chebyshev, rational, SVD/separated, or simple interpolatory matrix approximations in ω , with first- or second-order variation in the geometry amplitude ϵ of

$$c(\epsilon) = c_0 + \epsilon d.$$

Hold out both frequencies and geometry values.

Measure matrix-action error, receiver error, directional gradient error, effective frequency rank/order, maximum valid $|\epsilon|$, preparation work and predicted break-even count. This tests the mechanism without committing to analytic derivative implementation.

A strong **go** result would be: gradient errors below about 10^{-3} , forward receiver errors below the reference discretization budget, a shape-validity interval at least as large as several realistic trial steps from the current optimizer, and a frequency representation requiring only a small fraction of the full

frequency assemblies. As a provisional rule, if the preparation cost breaks even after fewer than roughly five full forward/gradient evaluations in the local region, the mechanism is worth developing.

The **stop condition** is especially important: if acceptable gradients require such a small $|\epsilon|$ that every accepted geometry update exits the surrogate region, or if adaptive frequency subdivision approaches one full matrix build per target frequency, the claimed broadband/geometry reuse is illusory.

If that experiment passes, the real research problem is to devise an error indicator that says when to rebuild. That, rather than a Taylor expansion itself, is where I would seek the paper's new mathematical mechanism.

The decisive test for reduced-basis/recycling

Hypothesis. Boundary traces and adjoints over the intended frequency band remain low-dimensional even after modest geometry changes.

Collect existing full-order solutions into

$$X = [x(c_1, \omega_1), \dots, x(c_s, \omega_q)]$$

and, ideally, collect adjoints in a second matrix. Compute the singular-value decay and construct U from only training frequencies/geometries. Test receiver fields and objective derivatives on held-out pairs.

No new BIE mathematics is required for this experiment.

A practical **go** criterion is that a basis dimension $r \lesssim 20\% - 25\%$ of the full trace dimension meets the receiver/gradient tolerance over held-out frequencies and at least one nontrivial geometry perturbation, *and* solve/factorization cost is a sufficiently large portion of the measured end-to-end runtime that reducing it matters. A useful end-to-end threshold would be an observed or realistically projected factor of two rather than an impressive reduced-solve timing hiding unchanged assembly.

Stop if r rises above roughly 40% of the full dimension, if the basis must be refreshed at nearly every geometry update, or if matrix/derivative assembly already consumes most of the runtime. In any of those cases, solution reduction is not the right primary bottleneck.

The inverse-validation gate

None of the three directions should receive an inverse-method claim solely from forward tests. Once a candidate passes its small diagnostic, perform one deliberately modest reconstruction:

four to six active geometry coefficients $\times$ several illuminations $\times$ roughly 8–16 frequencies.

Generate synthetic observations with a **more refined, independent Kress-Nyström discretization** than the inversion code, preferably at a geometry that is not exactly represented by the inverse truncation, and add a controlled noise level. That avoids the easiest inverse crime.

Compare reference and accelerated methods by accepted optimization steps, objective history, geometry error, receiver residual, gradient directional checks, and, most importantly,

$$T_{\text{total}} = T_{\text{geometry}} + T_{\text{assembly}} + T_{\text{factor}} + T_{\text{solves}} + T_{\text{receivers}} + T_{\text{derivatives}} + T_{\text{rebuild/update}}.$$

Report memory as well. Frequency count and transmitter count determine amortization, so both must be explicit. An accelerated kernel that causes more optimization iterations is not necessarily an accelerated inversion.

Staged research path and decision implications

The proposed path intentionally postpones every dependency-heavy commitment.

First, measure the existing pipeline rather than optimizing it. Establish accuracy-versus- (N, M) , then obtain separate timings for geometry processing, each Müller block, factorization, one right-hand side, all illumination solves, receiver evaluation, and one full gradient. This determines whether assembly, solves, or derivatives are actually the limiting resource. The literature shows that both matrix-frequency approximation and solution MOR can be successful, but they target different costs. ³⁶

Next, perform the post-processed modal-compressibility experiment. This is the highest-value decision gate because it simultaneously informs three ideas: hybrid Fourier assembly, adaptive trace/mode coupling, and whether direct Laurent coefficient work has a realistic payoff. It uses the trusted Kress code and therefore does not depend on proving a new quadrature correct.

If modal forward and derivative matrices are strongly compressible, implement the **hybrid** Galerkin assembler—not the fully Laurent one. Establish equivalence at the continuous-discretization level by independently refining both the hybrid Galerkin discretization and the Nyström reference. Remember that P^*A_NP is a diagnostic/control, not the definition of the final Galerkin operator.

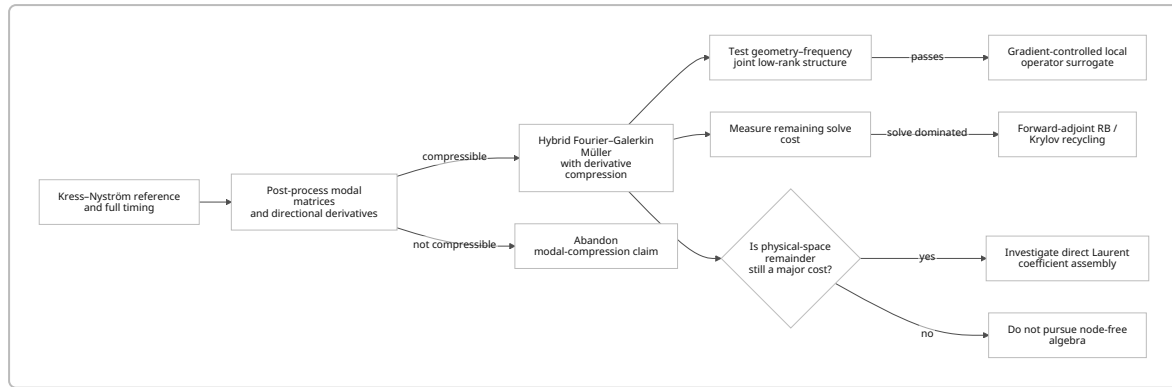
Once the hybrid assembler is reliable, determine whether its smooth coefficient data also exhibit low frequency rank. This naturally opens the second direction without duplicating infrastructure: the same modal blocks can become the objects in the geometry–frequency surrogate.

Only after real timing data show that full solves remain important should reduced-basis/Krylov recycling be added. It can then complement operator compression rather than conceal an assembly bottleneck.

I would postpone **full direct Laurent assembly** until three empirical facts are established: the modal interaction set is indeed sparse/compressible at the relevant ka ; derivative matrices retain that structure; and smooth-remainder quadrature/FFT work is large enough that eliminating it could materially affect end-to-end time. Without those facts, coefficient algebra is a mathematical detour rather than a decision-driven research step.

Likewise, multi-object scattering matrices should wait until the scientific problem actually contains multiple inclusions. Their reuse mechanism is attractive, but scattering-matrix/multipole coupling for arbitrary smooth particles is already established in nearby wave systems. ³⁵ Layered-ground GPR and three-dimensional electromagnetics would introduce larger physical and numerical changes still, so they should not be used to create novelty before the homogeneous full-space problem yields a clear methodological result.

The resulting dependency chain is:



The central research decision is therefore:

Work next on the structure of the modal operator and its shape derivative, not on a full new assembler. The new knowledge this could produce is whether the universal singular Fourier structure plus boundary smoothness creates a quantitatively exploitable, geometry-dependent coupling law for a transmission Müller system—and whether that law survives exactly the operation inverse scattering cares about: shape differentiation.

If it does, there is a credible numerical-analysis paper with a clean explanation: **“singular physics is structured; geometry mixes modes through a smoother remainder; we only retain the couplings required for both fields and gradients.”** If it does not, you will learn this before investing in delicate Laurent/Hankel coefficient algebra.

The second experiment should then ask whether the already-compressed/modal operator can also be reused in a local geometry–frequency neighborhood. That could become the higher-payoff follow-on paper. Generic RB/Krylov recycling remains a useful control and engineering accelerator but, on the current literature, is not the strongest primary novelty.

Verified references and search boundary

The literature search was carried through **17 September 2026**. Particularly important recent items include the March/April 2026 SIAM shape-Taylor papers and the September 2026 two-dimensional TM electromagnetic sensitivity paper, so the novelty assessment should not be based on a pre-2025 view of shape sensitivity. ³⁷

Foundational spectral and modal BIE work. R. Kress, “Boundary integral equations in time-harmonic acoustic scattering,” *Mathematical and Computer Modelling* 15, 229–243 (1991), compares Nyström, collocation and Galerkin trigonometric schemes and logarithmic singular quadratures for smooth two-dimensional Helmholtz boundaries. ¹² H. Cai and Y. Xu, “A Fast Fourier–Galerkin Method for Solving Singular Boundary Integral Equations,” *SIAM Journal on Numerical Analysis* 46, 1965–1984 (2008), develops $O(n \log n)$ -scale Fourier matrix compression for a smooth compact-kernel component. ¹³ T. Betcke, J. Phillips and E. A. Spence, “Spectral decompositions and nonnormality of boundary integral operators in acoustic scattering,” *IMA Journal of Numerical Analysis* 34, 700–731 (2014; online 2013), analyzes circle/ellipse/general-smooth-domain Fourier spectral structure. ⁸ J. Pinto, R. Aylwin and C. Jerez-Hanckes, “Fast solver for quasi-periodic 2D-Helmholtz scattering in layered media,” *ESAIM: M2AN* 55, 2445–2472 (2021), gives a rigorous Fourier spectral Galerkin transmission scheme. ¹⁴

Shape derivatives and geometry perturbations. M. Costabel and F. Le Louër, “Shape derivatives of boundary integral operators in electromagnetic scattering. Part I,” *Integral Equations and Operator Theory* 72, 509–535 (2012), establishes broad operator shape differentiability results. ³⁸ G. Bao, H. Ma, J. Lai and J. Li, “Shape Taylor Expansion for Wave Scattering Problems,” *SIAM Journal on Applied Mathematics* (2026), develops high-order shape recurrences for acoustic/electromagnetic and transmission problems. ³⁹ G. Bao, J. Lai and H. Ma, “Computation of Shape Taylor Expansions for Scattering Problems in Two Dimensions,” *SIAM Journal on Scientific Computing* 48, A448–A473 (2026), gives the two-dimensional computational treatment including transmission. ⁴⁰

Frequency reuse and reduced models. S. Dirckx, D. Huybrechs and K. Meerbergen, “Frequency Extraction for BEM Matrices Arising From the 3D Scalar Helmholtz Equation,” *SIAM Journal on Scientific Computing* (2022), is a close matrix-level frequency-reuse precedent. ¹⁹ D. Panagiotopoulos, W. Desmet and E. Deckers, “Parametric model order reduction for acoustic boundary element method systems through a multiparameter Krylov subspaces recycling strategy,” *International Journal for Numerical Methods in Engineering* 123, 5546–5569 (2022), treats multiparameter BEM reduction. ³⁰ M. Fares, J. S. Hesthaven, Y. Maday and B. Stamm, “The reduced basis method for the electric field integral equation,” *Journal of Computational Physics* 230, 5532–5555 (2011), is an important electromagnetic many-query RB precedent. ⁴¹ Y. Shi et al., “Reduced-basis boundary element method for fast electromagnetic field computation,” *JOSA A* 34, 2231–2242 (2017), applies RB-BEM to parameterized dielectric scatterers. ⁴² S. K. Baydoun et al., “A greedy reduced basis scheme for multifrequency solution of structural acoustic systems,” *International Journal for Numerical Methods in Engineering* 121, 187–200 (2020), gives a particularly useful frequency-sweep benchmark. ²⁹

Closest recent electromagnetic/shape-optimization overlaps. L. Chen et al., “Reduced order isogeometric boundary element methods for CAD-integrated shape optimization in electromagnetic scattering,” *Computer Methods in Applied Mechanics and Engineering* 419, 116654 (2024), combines electromagnetic frequency decoupling, SOAR MOR, IGABEM and multi-frequency shape optimization. ²⁰ C. Liu et al., “Isogeometric boundary element method analysis for dielectric target shape optimization in electromagnetic scattering” (2024), treats two-dimensional dielectric multi-frequency/wide-angle shape optimization. ²⁴ J. Shen et al., “Isogeometric boundary element analysis of sensitivity in TE-polarized electromagnetic scattering from dielectric bodies,” *Engineering Analysis with Boundary Elements* 181, 106549 (2025), derives and finite-difference-validates geometry sensitivities for two-dimensional dielectric scattering. ²² Y. Guo et al., “Isogeometric boundary element sensitivity analysis for TM-polarized electromagnetic scattering,” *Engineering Analysis with Boundary Elements* 190, 106839 (September 2026), is the most recent particularly close polarization/sensitivity precedent found in this search. ²³

Reuse under geometry change. Y. Zhang and A. Gillman, “A fast direct solver for boundary value problems on locally perturbed geometries,” *Journal of Computational Physics* 356, 356–371 (2018), and their 2021 extended-system follow-up demonstrate solver reuse when geometric modifications induce localized/low-rank matrix changes. ⁷ These are valuable controls but should not be assumed to transfer to global Fourier shape updates.

Coefficient-domain and multi-object precedents. M. Beceanu, J. Hong, H.-K. Kwon and M. Lim develop an infinite-dimensional Helmholtz layer-potential matrix through conformal mapping in work on planar Laplacian eigenvalues, demonstrating that coefficient-domain layer-potential algebra is a real prior mechanism, although in a substantially different problem. ⁴ J. Lai and P. Li, “A Framework for Simulation of Multiple Elastic Scattering in Two Dimensions,” *SIAM Journal on Scientific Computing* 41, A3276–A3299 (2019), combines local incident expansions, multipole scattered expansions and per-particle scattering matrices for arbitrary smooth particles. ⁹

Time-domain alternative. The 2026 work “High-Order Nyström/Convolution-Quadrature solution of time-Domain scattering from closed and open Lipschitz boundaries...” illustrates that modern CQ pipelines reduce time-domain scattering to families of complex-wavenumber Laplace-domain BIEs and can be paired with high-order Nyström methods. ¹⁰ It is technically strong prior art, but nothing found in this search gives CQ an obvious advantage sufficient to justify abandoning the present frequency-domain Müller backend for the stated inverse-scattering objective.

The main unresolved novelty question is therefore narrow rather than vague: **does the smooth remainder of the Fourier–Galerkin Müller operator, together with its Fourier-geometry derivatives, possess enough structured modal decay/compressibility to support a rigorously or adaptively controlled inverse solver?** That is the question with the best combination of falsifiability, mathematical content, implementation containment, and defensible distance from the closest 2024–2026 work.

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